Antimicrobial Therapy for Legionnaires' Disease

ANDREW TRAN
FOURTH YEAR STUDENT PHARMACIST
TEXAS A&M COLLEGE OF PHARMACY
ROTATION: INSTITUTIONAL
PRECEPTOR: MELANIE MOSS
12/15/2016

Objectives

- By the end of this presentation, the audience should be able to:
  - Identify risk factors associated with Legionnaires’ Disease
  - Describe appropriate tests for Legionnaires’
  - Recognize appropriate antimicrobial regimens in treatment

Patient Case

- CC: Weakness, Altered Mental Status
- HPI: K.W. is a 52-year-old male who presents to the ED on 11/18 with weakness and altered mental status. Upon arrival, he was found to have SOB, wheezing, numerous lab anormalities, and possible sepsis.
- Admitted to ICU for further evaluation

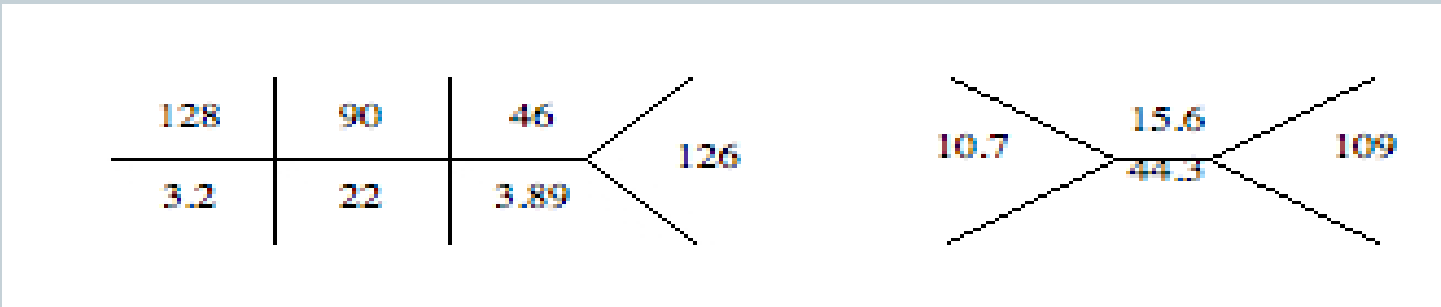
Patient Case

- PMH:
  - Alcohol abuse
    - 2-3 whiskeys per day
  - Anxiety
  - Depression
  - Hypertension
  - Tobacco Abuse
    - 50 pack-year history

Medication	Sig
Metoprolol tartrate 50 mg tablet	Take 1 tablet by mouth 2 times daily

Patient Case

- Initial Labs/Vitals/Cultures:



  - Wt: 87 kg, Ht: 69 inches
  - BP 126/87, **HR 142** , Temperature 99.7 , **RR 22**, **SPO2 88%**
  - Allergies: None
  - Gram stain negative, urine culture negative, blood, sputum, and CSF cultures pending.
  - Diagnosed with sepsis possibly due to pneumonia and suspected meningitis.

Lumbar Puncture Results

Component	Latest Ref Rng	11/18/2016
SOURCE, CSF	NA	CSF TUBE #3
VOLUME, CSF	mL	1.5
CLARITY, CSF	NA	CLEAR
COLOR, CSF	NA	COLORLESS
RBC, CSF	/mL	0
WBC COUNT, CSF	0 - 5 /mL	29 (CH)
SEGMENTED NEUTROPHILS, CSF	0 - 7 %	5
LYMPHS, CSF	40 - 80 %	69
MONO, CSF	15 - 45 %	26
DIFFERENTIAL BASED ON, CSF	CELLS NA	100

Patient Case

Day 1
- ED:
  - Start azithromycin 500 mg IV X 1 dose
  - Start ceftriaxone 2 G IV X 1 dose
- ICU:
  - d/c azithromycin
  - Continue ceftriaxone 2 g IV Q 12 H to cover for meningitis
  - Start vancomycin 1750 mg IV loading dose (20 mg/kg) + Zosyn 3.375 g IV QID
  - LP shows elevated WBC
  - Order blood, sputum, and CSF cultures, viral PCR

Day 2
- d/c Zosyn
- Start vancomycin 1000 mg IV q 36 H
- Start ampicillin 2 g IV q 8 H to cover for meningitis
- Start metronidazole 500 mg IV q 8 H

Day 3
- Cultures still negative; MRI negative
- CSF is not consistent with bacterial meningitis
- d/c ampicillin
- Start Unasyn 3 g IV q 6 H

- Day 4
  - ID consulted
  - d/c Unasyn
  - Start doxycycline 100 mg PO BID for 10 days to cover for environmental exposures
  - Ordered urinary antigen test, Hep C antibody, HIV antibody, leptosporosis antigen, crypotococcal antigen, typhus, bartonella and Q-fever serology

Patient Case

Day 5:
- Cultures still remain negative
- Fever has resolved
- Respiratory status has improved

Day 6
- Urinary Antigen tests positive for *Legionella pneumophila*
- Discontinue all active antibiotics
- Initiate **levofloxacin 750 mg IV q 24 H**
- Patient was extubated without complications

Legionnaires' Disease

- INTRODUCTION
- EPIDEMIOLOGY
- ETIOLOGY
- RISK FACTORS
- DIAGNOSIS
- TREATMENT

Legionnaires' Disease

- Legionnaires' Disease is a type of atypical pneumonia caused by bacteria of the *Legionella* species.
- The organism was first recognized in 1976 during an outbreak at an American Legion Convention in Philadelphia.

https://upload.wikimedia.org/wikipedia/en/9/93/AmerLegion_color_Emblem.jpg

Yu, Victor L., MD, and Nieves S. Galindo, MD. "Epidemiology and pathogenesis of Legionella infection." *UpToDate*. N.p., 21 J
2016. Web.

Epidemiology

- Legionnaires' disease accounts for approximately 1-9% of cases of community-acquired pneumonia (CAP).
- *Legionella pneumophila* serogroup 1 accounts for 70% of cases of Legionnaires' disease.
- Mortality rate of 15-25%
- All cases must be reported to state health department

Legionnaires Disease (2016). In Ucentral Unbound Medicine: John Hopkins ABX Guide. Database. Retrieved from <http://www.unboundmedicine.com.lib-ezproxy.tamu.edu:2048//view/Johns_Hopkins_ABX_Guide/540307/all/Legionnaires'_Disease>

Etiology

- *Legionella* thrives in man-made aquatic environments with higher temperatures.

- Sources of Legionella include cooling towers, swimming pools, water systems, hot springs, or parts of air-conditioning systems of large buildings.

http://www.achrnews.com/ext/resources/2012/07-09-12/S-Freije-IMG-6416.jpg

Yu, Victor L., MD, and Nieves S. Galindo, MD. "Epidemiology and pathogenesis of Legionella infection." *UpToDate*. N.p., 21 J
2016. Web.

Risk Factors

Table 8. Epidemiologic conditions and/or risk factors related to specific pathogens in community-acquired pneumonia.

Condition	Commonly encountered pathogen(s)
Alcoholism	*Streptococcus pneumoniae*, oral anaerobes, *Klebsiella pneumoniae*, *Acinetobacter* species, *Mycobacterium tuberculosis*
COPD and/or smoking	*Haemophilus influenzae*, *Pseudomonas aeruginosa*, *Legionella* species, *S. pneumoniae*, *Moraxella cararrhalis*, *Chlamydophila pneumoniae*
Aspiration	Gram-negative enteric pathogens, oral anaerobes
Lung abscess	CA-MRSA, oral anaerobes, endemic fungal pneumonia, *M. tuberculosis*, atypical mycobacteria
Exposure to bat or bird droppings	*Histoplasma capsulatum*
Exposure to birds	*Chlamydophila psittaci* (if poultry: avian influenza)
Exposure to rabbits	*Francisella tularensis*
Exposure to farm animals or parturient cats	*Coxiella burnetti* (Q fever)
HIV infection (early)	*S. pneumoniae*, *H. influenzae*, *M. tuberculosis*
HIV infection (late)	The pathogens listed for early infection plus *Pneumocystis jirovecii*, *Cryptococcus*, *Histoplasma*, *Aspergillus*, atypical mycobacteria (especially *Mycobacterium kansasii*), *P. aeruginosa*, *H. influenzae*
Hotel or cruise ship stay in previous 2 weeks	*Legionella* species
Travel to or residence in southwestern United States	*Coccidioides* species, *Hantavirus*
Travel to or residence in Southeast and East Asia	*Burkholderia pseudomallei*, avian influenza, SARS
Influenza active in community	Influenza, *S. pneumoniae*, *Staphylococcus aureus*, *H. influenzae*

Mandell LA et al: Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis* 44 Suppl 2:S27, 2007

Diagnosis

- If Legionnaires' is suspected, both a **urinary antigen test and a respiratory culture** should be ordered.
  - Clinical suspicion: severe or life-threatening pneumonia, especially in patients with risk factors
- Urine antigen tests only for *L. pneumophila*
  - Detects 70% of such cases but nearly 100% specific
- Sputum culture on special media (BCYE) and requires > 3 days for growth

Legionnaires Disease (2016). In Ucentral Unbound Medicine: John Hopkins ABX Guide. Database. Retrieved from <http://www.unboundmedicine.com.lib-ezproxy.tamu.edu:2048//view/Johns_Hopkins_ABX_Guide/540307/all/Legionnaires'_Disease>

Criteria for Severe CAP

Table 4. Criteria for severe community-acquired pneumonia.

Minor criteria[a]	
	Respiratory rate[b] $\geqslant$30 breaths/min
	PaO_2/FiO_2 ratio[b] $\leqslant$250
	Multilobar infiltrates
	Confusion/disorientation
	Uremia (BUN level, $\geqslant$20 mg/dL)
	Leukopenia[c] (WBC count, <4000 cells/mm^3)
	Thrombocytopenia (platelet count, <100,000 cells/mm^3)
	Hypothermia (core temperature, <36°C)
	Hypotension requiring aggressive fluid resuscitation
Major criteria	
	Invasive mechanical ventilation
	Septic shock with the need for vasopressors

NOTE. BUN, blood urea nitrogen; PaO_2/FiO_2, arterial oxygen pressure/fraction of inspired oxygen; WBC, white blood cell.

[a] Other criteria to consider include hypoglycemia (in nondiabetic patients), acute alcoholism/alcoholic withdrawal, hyponatremia, unexplained metabolic acidosis or elevated lactate level, cirrhosis, and asplenia.

[b] A need for noninvasive ventilation can substitute for a respiratory rate >30 breaths/min or a PaO_2/FiO_2 ratio <250.

[c] As a result of infection alone.

Mandell LA et al: Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis* 44 Suppl 2:S27, 2007

Antimicrobial Therapy

Inpatient, ICU treatment. The following regimen is the minimal recommended treatment for patients admitted to the ICU.

20. A β-lactam (cefotaxime, ceftriaxone, or ampicillin-sulbactam) **plus** either azithromycin (level II evidence) or a fluoroquinolone (level I evidence) (strong recommendation) (For penicillin-allergic patients, a respiratory fluoroquinolone and aztreonam are recommended.)

Mandell LA et al: Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis* 44 Suppl 2:S27, 2007

Antimicrobial Therapy

Table 9. Recommended antimicrobial therapy for specific pathogens.

Organism	Preferred antimicrobial(s)	Alternative antimicrobial(s)
Streptococcus pneumoniae		
Penicillin nonresistant; MIC <2 μg/mL	Penicillin G, amoxicillin	Macrolide, cephalosporins (oral [cefpodoxime, cefprozil, cefuroxime, cefdinir, cefditoren] or parenteral [cefuroxime, ceftriaxone, cefotaxime]), clindamycin, doxycyline, respiratory fluoroquinolone[a]
Penicillin resistant; MIC ≥2 μg/mL	Agents chosen on the basis of susceptibility, including cefotaxime, ceftriaxone, fluoroquinolone	Vancomycin, linezolid, high-dose amoxicillin (3 g/day with penicillin MIC ≤4 μg/mL)
Haemophilus influenzae		
Non–β-lactamase producing	Amoxicillin	Fluoroquinolone, doxycycline, azithromycin, clarithromycin[b]
β-Lactamase producing	Second- or third-generation cephalosporin, amoxicillin-clavulanate	Fluoroquinolone, doxycycline, azithromycin, clarithromycin[b]
Mycoplasma pneumoniae/Chlamydophila pneumoniae	Macrolide, a tetracycline	Fluoroquinolone
Legionella species	Fluoroquinolone, azithromycin	Doxycyline

Mandell LA et al: Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis* 44 Suppl 2:S27, 2007

Antimicrobial Therapy

- Preferred: a fluoroquinolone or macrolide
  - **Levofloxacin** 750mg IV q24h then 750mg PO daily x 10d
  - **Moxifloxacin** 400mg IV q24h, then 400mg PO daily x 10d
  - **Azithromycin** 500mg IV, then 500mg PO daily x 7-10d
- **Rifampin** 300mg IV or PO twice daily used in combination with **moxifloxacin**, **levofloxacin** or **azithromycin**. Benefit of combination therapy is unclear, used mainly w/ severe illness.

Legionnaires Disease (2016). In Ucentral Unbound Medicine: John Hopkins ABX Guide. Database. Retrieved from <http://www.unboundmedicine.com.lib-ezproxy.tamu.edu:2048//view/Johns_Hopkins_ABX_Guide/540307/all/Legionnaires'_Disease>

Sanford Guide Activity Spectra

Activity Spectra

Legend

Activity Spectra	Fluoroquinolone						Macrolides				Tetracycline		
	Ciprofloxacin	Ofloxacin	Levofloxacin	Moxifloxacin	Gemifloxacin	Gatifloxacin	Erythromycin	Azithromycin	Clarithromycin	Telithromycin	Doxycycline	Minocycline	Tigecycline
F. tularensis	++	?	?	?	?	?	0	?	?	?	++	+	?
H. ducreyi	++	?	+	?	?	?	++	++	?	?	0	0	0
H. influenzae	+	+	++	++	+	+	0	+	+	+	+	+	+
Kingella sp.	+	+	++	?	?	?	0	?	?	?	+	+	+
K. granulomatis	++	?	?	?	?	?	++	++	+	0	++	+	?
Legionella sp.	++	++	++	++	+	+	++	++	++	+	+	+	+
Leptospira sp.	+	+	+	+	+	+	+	++	+	?	++	+	?
M. catarrhalis	+	+	+	+	+	+	+	+	+	+	+	+	?
N. gonorrhoeae	±	±	±	0	+	0	0	+	0	0	±	±	?
N. meningitidis	?	?	?	?	?	?	0	0	0	0	0	0	0

DN Gilbert, RC Moellering Jr, GM Eliopoulos, HF Chambers, MS Saag (Eds.) The *Sanford Guide* to Antimicrobial Therapy 2016. 46th ed.

Previous Trials

Study	Objective	Methods	Results	Citation
Antimicrobial chemotherapy for Legionnaires disease: levofloxacin versus macrolides. (2005)	Compare clinical response of patients with Legionella pneumonia treated with levofloxacin with that of patients treated with macrolides (erythromycin or clarithromycin) and evaluate the role of rifampicin combined with levofloxacin	• Observational, prospective, nonrandomized study involving 292 patients • Compared antibiotic regimens (macrolides vs levofloxacin)	• All patients were cured • Levofloxacin associated with fewer complications and shorter mean hospital stays • Addition of rifampicin to the treatment regimen for patients receiving levofloxacin for severe pneumonia provides no benefit	Clin Infect Dis. 2005 Mar 15;40(6):800-6. Epub 2005 Feb 17.

Previous Trials

Study	Objective	Methods	Results	Citation
Macrolides versus quinolones in Legionella pneumonia: results from the Community-Acquired Pneumonia Organization international study. (2010)	Compare time to clinical stability and length of hospital stay in patients with Legionella pneumonia who were treated with levofloxacin compared to those treated with macrolide (azithromycin) therapy.	Retrospective analysis of 39 patients with Legionnaires' disease from Community-Acquired Pneumonia Organization data base.	• Time to clinical stability and length of hospital stay were not statistically different between macrolide and levofloxacin groups.	Int J Tuberc Lung Dis. 2010 Apr;14(4):495-9.

Back to the Patient Case

K.W. Clinical Course

ED Visit: azithromycin + ceftriaxone

- Start ampicillin + metronidazole
- d/c Zosyn

- d/c ampicillin
- - start Unasyn

- Cultures negative
- Fever resolved

- d/c levofloxacin IV
- Start levofloxacin 750 mg PO daily

Day	1	2	3	4	5	6	7	8	9	10	11
WBC	10.7	5.4	4.4	3.7		5.5	7.1	7.7			7.6
SCr	3.89	3.25	3.37	2.18	1.83	1.65	1.4	1.21			1.15

- Pt was intubated
- LP showed elevated WBC
- d/c azithromycin
- switch to Zosyn + Vancomycin
- Ordered blood, sputum, urine, CSF, and gram stain.

- ID consulted
- Ordered urinary antigen test, Hep C antibody, HIV antibody, leptosporosis antigen, crypotococcal antigen, typhus, bartonella and Q-fever serology
- d/c Unasyn
- Start doxycycline.
- Continue ceftriaxone and metronidazole

- Urine antigen test positive for legionella
- d/c all current antibiotics
- Start levofloxacin 750 mg IV q 24 H
- Pt was extubated w/o complications

- ID consulted
- Start metronidazole 500 mg PO TID for sinusitis

Discharged with 14 more days of levofloxacin and 7 more days of metronidazole

K.W. Antibiotic Therapy Overview

Day	1	2	3	4	5	6	7	8	9	10	11
Ampicillin IV		■	■	■							
Ampicillin-sulbactam (Unasyn®) IV				■							
Azithromycin IV	■										
Ceftriaxone IV	■	■	■	■	■						
Doxycycline PO					■	■					
Levofloxacin IV						■	■	■			
Levofloxacin PO									■	■	■
Metronidazole IV		■	■	■	■						
Metronidazole PO										■	■
Piperacillin-tazobactam (Zosyn®) IV	■	■									
Vancomycin IV	■	■	■	■	■						

Assessment of Management

- Initial empiric therapy for CAP: ceftriaxone + azithromycin
- Prior to antigen test, broad antimicrobial therapy
- Urinary antigen test on day 6
- Pathogen specific therapy for *Legionella pneumophila*: levofloxacin
- Long duration of therapy for levofloxacin (20 days total)

Key Points

- Legionnaires' is a rare and deadly form of pneumonia
- Suspected in severe or life-threatening pneumonia
- Urinary antigen test: very specific and fast
- Respiratory culture:
  - specific media required
  - 3 days required for growth
- First-line treatment consists of fluoroquinolone or macrolide therapy

Questions?